

BAAZ JKBOSE



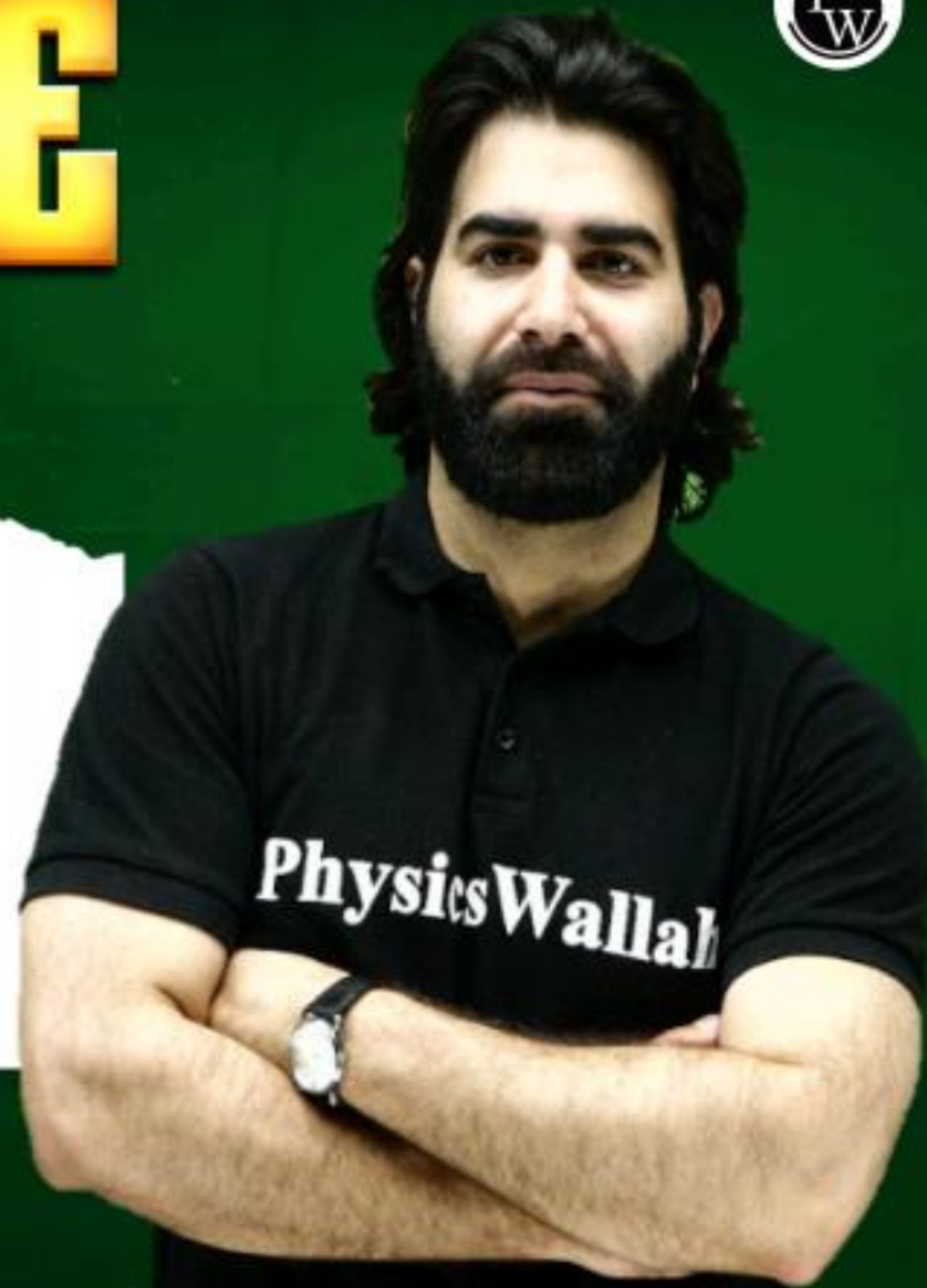
2026 CLASS-11TH

Excretory Products and
Their Elimination

LECTURE-1

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 INTRODUCTION, Types of nitrogenous wastes

Previous Chapter (few Topic)

- Regulation of Heart Beat / Cardiac Activity

① Neural Regulation.

- There is Cardiac Centre in the Medulla of Brain.

[It uses Autonomic Nervous System]
ANS
SNS PNS

Cardiac centre;

② Cardiac Accelerator Centre.

- It increase Heart Rate
- It sends message via Sympathetic Nervous System.

③ Cardiac Inhibitory Centre:

- It decreases Heart Rate
- It sends message via Parasympathetic Nervous System

→ Some Hormones also increase Heart Rate, mainly.

④ Adrenaline (Adrenal Gland).

⑤ Thyroxine (Thyroid Gland).

Disorders.

① Hypertension.



- ↳ It refers to increased Blood Pressure
- ↳ Blood pressure is the pressure exerted by the Blood on the walls of Blood vessels.
- ↳ It is Represented as Systolic Pressure (When Heart Contracts)
Diastolic Pressure (When Heart Relaxes).

↳ Mostly we Measure Pressure of Arteries

↳ Normal Blood Pressure = $\frac{120 \text{ mmHg}}{80 \text{ mmHg}}$ [Measured in mmHg]
(Average)

↳ Hypertension is when B.P is above $\frac{140}{90}$

② Atherosclerosis → ^{(Hardening).} Stiffening of Blood vessels due to Deposition of

- Fatty substance (cholesterol).
- Salts (Ca^{++}) etc.

→ It is Related to Life Style ..

[Ischemia - Less Blood Flow to Tissue]

③ Coronary Artery Disease (CAD) / Ischemic Heart Disease

→ It is atherosclerosis of Heart Arteries, due to which less Blood will Reach to Heart tissue

↳ It can lead to Heart Failure, Heart Attack and finally Cardiac Arrest.

④ Heart Failure → It is insufficient Pumping Action of Heart. [$<40\%$ pumping by ventricles].

→ Congestion of Chest is one of the Important symptom, that is why it is also k/as
(Congestive Heart Failure)

→ It is due to weakening of Heart Muscle.

⑤ Heart Attack / Myocardial Infarction (MI).
 Heart Muscle Death of Tissue.

↳ Damage / Death of Heart Muscles, due to Sudden Blood Blockage / Stoppage

⑥ Cardiac Arrest - It is complete end of activities of heart. (No-Beating)

④ Angina → Acute chest Pain.

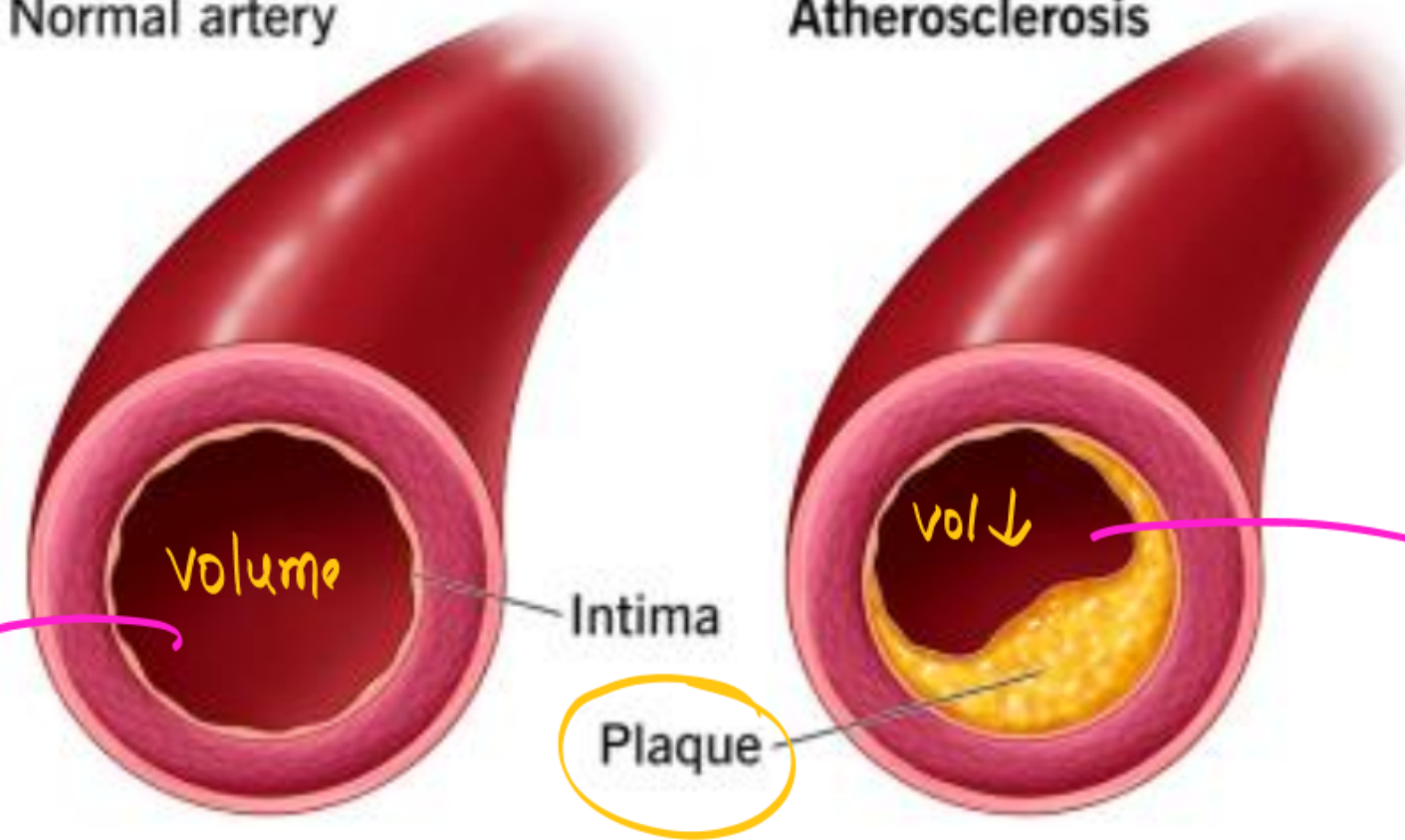
→ often due to low Blood Supply to Heart Tissue.

→ Like seen in Heart Failure / Heart Attack.

Atherosclerosis

Normal artery

Atherosclerosis



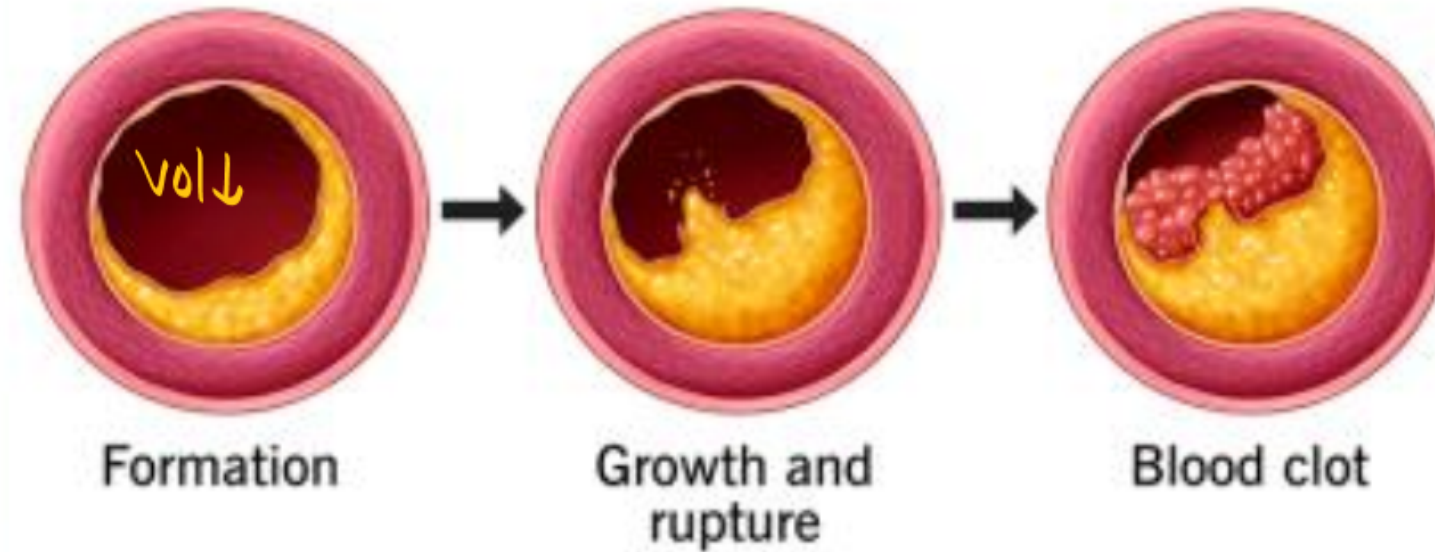
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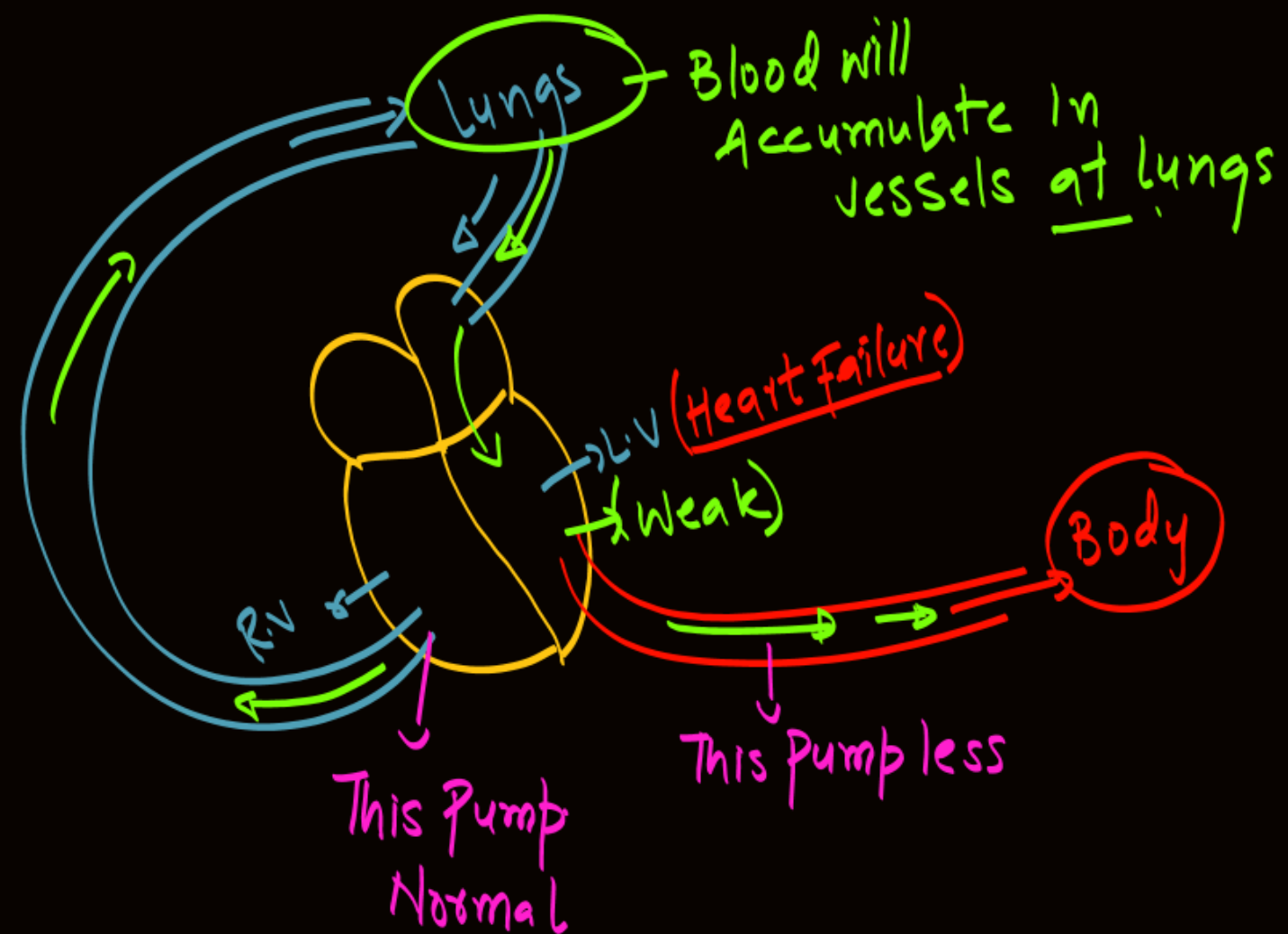
More Blood will Reach Tissue

20's, 30's, 40's
60, 70's

Less Blood will Reach to tissues (Ischemia)

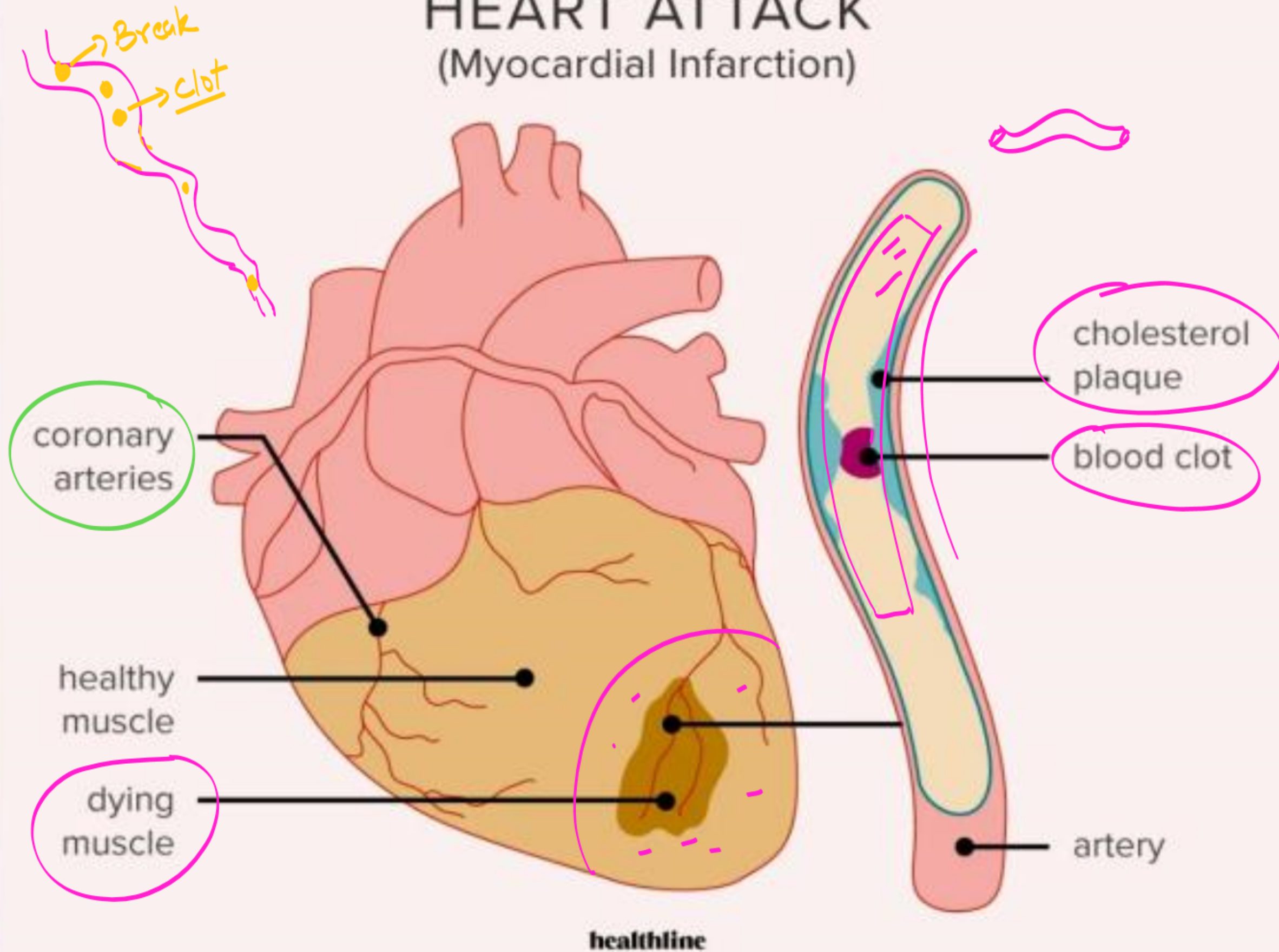
Progression





HEART ATTACK

(Myocardial Infarction)



New Chapter [Excretory products and their Elimination]

Excretion

It is Removal of waste / Excretory products from Body.

→ Wastes like

→ Nitrogenous wastes, ^(toxic) includes Ammonia, Urea, Uric Acid etc

→ Excess of H₂O, ions Na⁺, K⁺, Ca²⁺, PO₄⁻ etc (salts).

To Maintain Osmoregulation [Water & Salt Balance].

→ By kidneys

→ Bilirubin, Bilirubin [By Liver]

→ CO₂ etc [By Lungs]



a) Ammonia.



↳ It is from Deamination of Amino Acids, Nitrogenous Bases etc

↳ NH_3 (Ammonia) in Body is in the form of Ammonium ion



↳ NH_4^+ is Highly Soluble & Highly Toxic, and requires Most Amount of H_2O for Excretion.

↳ organism whose Main Waste is Ammonia → Ammoniotelic
(Process - Ammoniotelism)

e.g Bony Fishes, Aquatic Amphibians, Aquatic Insects.

Types of Nitrogenous wastes:

b) Urea ($\text{H}_2\text{N}-\text{CO}-\text{NH}_2$)

- Urea is Formed from Ammonia & CO_2 in Urea Cycle (Ornithine cycle) (Liver)

- Urea is less toxic than Ammonia and requires less H_2O than Ammonia.

- organism whose Main Waste is Urea - Ureotelic
(Process - Ureotelism)

e.g - Mammals (Humans).

- Terrestrial Amphibians (Frog)
- Cartilageneous Fishes

c) Uric Acid.

- Uric Acid is least toxic and least Soluble in H_2O .

- often precipitates in H_2O .

- organism producing it (Main Waste)

- Uricotelic

Process - Uricotelism.

e.g Reptiles (pellets).

Birds (Paste)

Terrestrial Insects.

Land Snails

Qno1) - What is the Main Nitrogenous waste in Humans?

- (a) Ammonia
- (b) Urea
- (c) Uric Acid
- (d) Creatinine

Qno2) Bony Fishes are

- (a) Ammoniotelic
- (b) Ureotelic
- (c) Uricotelic
- (d) None of these

QUESTIONS

Which of the following feature(s) is common to all those compounds found in acid soluble pool or filtrate?

- 1 They are called macromolecules or biomacromolecules.
- 2 The molecular weight is more than ten thousands Daltons or above.
- 3 They have molecular weight ranging from 18 to around 800 Da approximately.
- 4 Both A and B

HOMEWORK



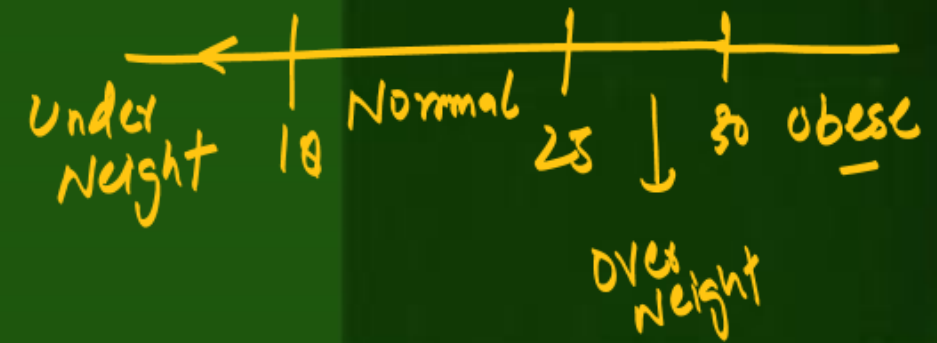
Gout.

||
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300 X

$$\text{BMI} = \frac{\text{Weight in kg}}{(\text{Height in M})^2} = ?$$

Body Mass Index.



Thank You